

An input image **227X227X3** is convolved with **100** filters having size = **21X21X3**, stride = **4** and zero padding = **1**. In the 2nd phase output is convolved with **50** filters having size = **7X7X100**, stride = **2** and zero padding = **0**. Find out the sizes of the outputs and parameter counts after each phase.

Answer:

1st Phase-

Input= 227X227X3

Output dimension= $(W-F+2P)/s + 1 = (227-21+2*1)/4 + 1 = 53$

Output size or neuron = 53X53X100. (Reduced Size X Number of Channels)

Number of weights = 21X21X3X100 [Filter Size, Number of Filters]

Number of Bias = 100

Total Parameter= 21X21X3X100 + 100=132400

2nd Phase-

Input=53X53X100

Output dimension= $(W-F+2P)/s + 1 = (53-7+2*0)/2 + 1 = 24$

Output size or Neuron = 24X24X50

Number of Weights = 7X7X100X50

Number of Bias = 50

Total Parameter= 7X7X100X50 + 50=245050

Show the first convolution operation using two matrix multiplications.

Explain what is represented by rows and columns of each matrix.

Answer:

Image matrix = $(21 * 21 * 3) * (53 * 53) = 1323 * 2809$

Number of rows (values in block) = $21 * 21 * 3$

Number of columns (blocks) = $53 * 53$

Filter matrix = $100 * 1323$

Number of rows (number of filters) = 100

Number of columns (values of a filter) = $21 * 21 * 3$

Output Matrix = Filter Matrix * Image Matrix = $(100 * 1323) * (1323 * 2809) = 100 * 2809$